BRFSS

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Abstract

This report describes a preprocessing pipeline for BRFSS survey data from 2022–2024 to create a modeling-ready dataset for predicting mental health (MENTHLTH) and life satisfaction (LSATISFY). The workflow includes multi-year data integration (initial 1.3M observations, 478 variables harmonized to 225 common columns), BRFSS-specific missing-code handling (e.g., converting 88 to 0 for health days), BMI scaling, balanced year-wise sampling (13,333 per year → 39,999 final observations), removal of identifiers, redundant/leakage variables, and features with >50% missingness. Categorical variables were encoded, and remaining missing values were median-imputed. The final dataset contains 39,999 observations and 67 predictor features, ready for modeling.

Overview

Behavioral Risk Factor Surveillance System (BRFSS) data from **2022, 2023, and 2024** were combined and preprocessed to create a consistent dataset suitable for statistical analysis and machine-learning modeling. The primary prediction target was **MENTHLTH**, which represents the number of days during the past 30 days that a respondent reported their mental health was not good. **LSATISFY** (satisfaction with life) was retained as a secondary outcome variable.

The preprocessing pipeline included multi-year data integration, variable harmonization, BRFSS-specific missing-value processing, balanced sampling across years, feature selection, leakage prevention, categorical encoding, and missing-value imputation.

1. Data Loading

The BRFSS annual datasets were provided in SAS Transport (`.XPT`) format. Files for 2022, 2023, and 2024 were detected and loaded automatically using `pd.read_sas(file_path, format='xport')`. A `YEAR` variable was added to identify the source year for each observation.

After combining the three datasets, the initial merged dataset contained:

Dataset characteristic	Value
Years	2022 - 2023 - 2024
Initial observations	1,336,125
Columns before harmonization	478
Variables common to all three years	224

Dataset characteristic	Value
Columns after harmonization, including YEAR	225

Only variables appearing in **all three annual datasets** were retained. This ensured that each selected feature had a consistent definition and availability across the study period.

2. Data preprocessing

1.1. BRFSS Missing-Value Processing

BRFSS uses numeric response codes to represent responses such as *Don't know/Not sure*, *Refused*, and *Missing/Unknown*. Therefore, these codes could not be treated as ordinary numerical values.

For most variables, codes such as `7`, `8`, `9`, `77`, `88`, `99`, `777`, `888`, `999`, and their extended forms were treated as missing values and converted to `NaN`.

However, special handling was required for variables in which values such as 7, 8, 9, and 88 can represent legitimate quantities. These included variables related to:

- number of unhealthy days,
- household and child counts,
- age,
- alcohol consumption,
- cigarette consumption,
- income and education categories, and
- time/frequency measures.

For the health-day variables `MENTHLTH`, `PHYSHLTH`, and `POORHLTH`, a value of **88 was converted to 0 days**, while values 7, 8, and 9 were preserved as valid numbers of days. Extended missing codes were converted to `NaN`.

This variable-specific treatment prevented valid numerical responses from being incorrectly classified as missing data.

1.2. Target Cleaning

The primary target variable was:

MENTHLTH — number of mentally unhealthy days during the previous 30 days (0–30 days).

`LSATISFY` was retained as a secondary outcome variable. Observations with missing values for either `MENTHLTH` or `LSATISFY` were removed before subsequent sampling and modeling.

1.3. Missing Value Assessment

Before imputation, we examined missingness across all candidate predictors. Features with more than 50% missing values were removed (see 1.7). For the remaining features, the proportion of missing values ranged from 0% to 48%. We applied median imputation (see 1.10) because:

- The distributions of most continuous features (e.g., BMI, age) were skewed, making median more robust than mean.
- Median imputation preserves the central tendency without being overly influenced by outliers, which is critical for tree-based models.
- No systematic pattern of missingness was observed that would warrant more complex imputation (e.g., MICE).

We confirmed that missingness was not strongly correlated with the target (MENTHLTH / LSATISFY), reducing the risk of imputation-induced bias.

1.4. BMI Transformation

The BRFSS `_BMI5` variable is stored as BMI multiplied by 100. A new `BMI` variable was therefore calculated as:

```
df['BMI'] = df['_BMI5'] / 100.0
```

For example, `_BMI5 = 2500` was converted to `BMI = 25.0`.

Among observations with available BMI before final sampling, the median BMI was **27.44 kg/m^2** and the mean was approximately **28.59 kg/m^2**.

1.5. Balanced Sampling Across Years

Because BRFSS contains different numbers of respondents in each year, equal sampling was used to prevent years with larger datasets from having disproportionate influence on the analysis.

The target total of ~40,000 observations was chosen to balance computational efficiency and statistical power. With 13,333 per year, we maintain adequate representation of each year while keeping the dataset manageable for extensive hyperparameter tuning and cross-validation.

After target-related cleaning, the available numbers of observations were:

Year	Available observations
2022	246,845
2023	220,510
2024	198,750

The desired total sample size was approximately **40,000 observations**. Therefore, **13,333 respondents were randomly selected from each year**, resulting in **39,999 observations**.

A fixed random seed (`random_state=42`) was used to make the sampling reproducible. The observations from the three years were then combined and randomly shuffled.

1.6. Feature Removal and Selection

Several categories of variables were removed before modeling.

1.6.1. Identification and Survey-Design Variables

Variables that primarily identify respondents or describe the BRFSS sampling/interview process were excluded. These included variables such as `SEQNO`, `_PSU`, `_STSTR`, interview date variables, `DISPCODE`, and `LANDLINE`. The target variables were also removed from the predictor matrix to prevent them from being used as model inputs.

1.6.2. Redundant Variables

Variables containing duplicate or highly similar representations of the same information were removed. Examples included multiple representations of:

- age,
- height,
- weight, and
- BRFSS survey weighting variables.

For example, `_AGE80` was retained as the selected age representation while other derived age-group variables were excluded.

1.6.3. Questionnaire-Administration Variables

Variables primarily describing questionnaire administration rather than respondent characteristics were removed, including:

`QSTLANG`, `YEAR`, `QSTVER`, and `CTELNUM1`.

1.6.4. Highly Correlated Variables

For groups of variables representing similar information, one representative feature was retained and correlated alternatives were removed. Examples included:

- smoking status,
- asthma status,
- physical-health measures,
- state,
- number of children,
- education,
- BMI, and
- sex.

For example, the newly calculated `BMI` variable was retained while `_BMI5`, `_BMI5CAT`, and `_RFBMI5` were removed. For another example, `_ASTHMS1` was retained while `ASTHMA3`, `_CASTHM1`, and `_LTASTH1` were removed.

1.7. Prevention of Target Leakage

Target leakage occurs when a predictor contains information directly derived from the outcome being predicted. Such variables can artificially improve model performance.

Because `MENTHLTH` was the primary target, `_MENT14D`, a CDC-derived variable based on mental-health days, was removed. `POORHLTH` was also excluded from the predictor matrix in the implemented preprocessing pipeline.

This step helps ensure that model performance reflects relationships between independent predictors and mental health rather than direct or transformed versions of the target.

1.8. Removal of Features with Extensive Missingness

The percentage of missing observations was calculated independently for each candidate predictor. Features with **50% or more missing values** were excluded.

Only variables with a missing fraction below 0.50 were retained. Following the complete feature-selection procedure, **67 predictor variables remained**.

1.9. Categorical Variable Encoding

Categorical variables stored as text (`object` datatype) were converted to integer codes using `pandas.factorize()`. Each unique category was assigned an integer value, allowing the data to be used by machine-learning algorithms that require numerical input.

These integer values represent category labels rather than numerical magnitudes.

1.10. Missing-Value Imputation

After feature selection, any remaining missing predictor values were filled using **median imputation**.

3. Final Preprocessed Dataset

The final predictor matrix contained:

Characteristic	Final value
Years represented	3
Observations	**39,999**

Characteristic	Final value
Observations per year	13,333
Predictor features	67
Primary target	MENTHLTH
MENTHLTH target range	0–30 days
Secondary target	LSATISFY
LSATISFY target range	1–4

The final feature matrix therefore had dimensions:

```
(39999, 67)
```

while the target vector contained 39,999 observations.

```
(39999, 1)
```

The distribution of `MENTHLTH` was strongly concentrated at zero: **24,228 of the 39,999 respondents reported zero mentally unhealthy days**, while responses extended through the full 0–30-day range.

4. Summary

The complete preprocessing workflow can be summarized as:

Raw BRFSS 2022–2024 data
→ merge annual datasets
→ retain variables common to all years
→ decode text/byte variables
→ convert BRFSS-specific missing codes to missing values
→ remove observations missing required outcomes
→ convert BMI to standard units
→ sample equally from each year
→ remove identifiers and survey-administration variables
→ remove redundant and highly correlated features
→ remove target-leakage variables
→ exclude features with ≥50% missing data
→ encode categorical variables
→ median-impute remaining missing values
→ produce final modeling dataset of **39,999 observations and 67 predictors**.

This preprocessing procedure produced a standardized and reproducible dataset designed to reduce missing-data problems, redundant information, temporal imbalance, and target leakage before model development.

5. Conclusion

Item	Value
Raw data years	2022–2024
Initial observations	1,336,125
Final observations	39,999
Initial features	478
Common features across years	225
Features after cleaning	67
Features used in modeling	40
Primary target	MENTHLTH (0–30 days)
Secondary target	LSATISFY (1–4)
Best model for MENTHLTH	XGBoost (MAE=4.14)
Best model for LSATISFY	LightGBM (MAE=0.415)